

PROJECT AND TEAM INFORMATION

Project Title

JEEVANROUTE: Emergency ambulance selection,hospital capability matching and route calculation

Team Name:	*CYPHERX* *DSCPP-III-2026-T194*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

- JEEVANROUTE is an emergency ambulance selection, hospital matching, and route calculation system developed in C++ to demonstrate data structures, algorithms, and object-oriented programming concepts.
- The system prioritizes critical patients using priority queues and identifies available ambulances based on emergency requirements.
- JEEVANROUTE selects suitable hospitals by checking essential facilities such as ICU beds, CT scans, blood supplies, and specialist doctors.
- The project uses graphs and pathfinding algorithms to calculate efficient ambulance routes and aims to improve coordination between ambulance operators and hospitals.
- The phase I, focuses on system architecture and module planning, followed by implementation of data management, emergency assessment, ambulance selection, hospital matching, and route calculation.

Updated Project Approach and Architecture (2 pts)

- We will continue using the same architecture as we did in Phase I; the project consists of 5 different components: Patient details, Emergency priorities, Ambulance management, Hospital selections, Route calculations.
- Initially we are going to plan how we can store patient information, ambulance details, hospital facilities and roads. from this information we can determine which ambulance should be sent to which hospital during an emergency.
- The project is c developed in C++ using various data structures such as priority queues, lists, and graphs. Priority queues will allow us to handle serious cases first, while graphs will help us find suitable routes for ambulances which will allow the ambulances to reach the hospital on time.
- The project work is divided into separate modules, including patient assessment, ambulance selection, hospital selection and route calculation. every team member will work on their assigned part before all the modules are combined.
- This approach makes the project easier to develop and test. Once the individual situation modules are ready , we will combine them into one system that aims to reduce delays and help patients reach hospitals with the facilities they need.

Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task Completed	Team Member
Finalized project architecture and module breakdown (Phase I)	All members
Set up shared GitHub repository with collaborator access	Abhishek Yadav
Established modular file structure for parallel development	Abhishek Yadav
Planned data structures for storing patient details, ambulance information, hospital facilities, and road network data	Archit singh
Designed the patient assessment and emergency prioritization modules using priority queues	Tushar Srivastava
Planned ambulance selection, hospital capability matching, and graph-based route calculation modules	Abhishek Yadav
Identified data structures and module interfaces for hospital pre-alerts and integration of the complete system	Aryan Singh

Challenges/Roadblocks (7 pts)

(Describe the challenges that you have faced or are facing so far and how you plan to solve them. Max 300 words).

- we have figured out a clean modular architectur for ambulance dispatch, hospital matching, and route planning .
- we have picked up the right C++ data structures to keep track of patient details, active ambulances, and hospital facilities without running into performance .
- prioritize emergency cases by severity so critical patients are handled first using a priority queue.
- selecting and matching hospitals based on real-time availability of ICU beds, working CT scanners, blood stock, and on-call specialists.
- selection of the road network as a weighted graph to run pathfinding algorithms like dijkstra for the fastest ambulance route.
- handling the edge cases when there are no available ambulances nearby or zero matching hospital beds for an emergency.
- During emergency edge cases with messy, incomplete, or invalid input data across patients, ambulances, and hospital records.
- we have kept everyone's code organized and merging all the separate C++ modules without breaking existing features .

Tasks Pending (7 pts)

Task Pending	**Team Member**
Finished up debugging and testing the patient assessment along with the emergency prioritization bits.	Archit Singh
Wrote the priority queue logic so critical emergency cases jump straight to the top of the stack.	Tushar Srivastava
Built the ambulance selection script keeping vehicle availability, GPS proximity, and patient needs in mind..	Archit Singh
Handled hospital matching based on open ICU beds, CT scan availability, blood stock, and specialist docs.	Abhishek Yadav
Set up graph-based routing to calculate the quickest paths for ambulances to dodge traffic.	Aryan Singh
Put together the pre-alert system to ping the selected hospital with patient vitals ahead of arrival.	Aryan Singh, Archit Singh
Hooked up the ambulance picker, hospital matcher, and route calculator into one smooth workflow.	Abhishek Yadav, Tushar Srivastava
Ran tests on a bunch of edge-case emergency scenarios and finally merged everything into one main C++ application.	All members

Project Outcome/Deliverables (2 pts)

(Describe what are the key outcomes / deliverables of the project. Max 200 words).

- By the end of Phase II, the JEEVANROUTE project architecture and modular structure have been planned, providing a foundation for developing and integrating the system components.
- We have planned the data management structure to store patient details, ambulance availability, hospital facilities, and information about road network.
- We have planned to use priority queues to handle critical emergencies first. The basic design for selecting an ambulance and matching a suitable hospital has also been prepared.
- We have planned to use graphs to represent the road network and find suitable routes for ambulances and the system will also include a feature to alert hospitals before the ambulance arrives.
- The remaining work includes developing the planned modules, connecting ambulance selection with hospital matching and route calculation, and testing the complete system. We will focus on these tasks in Phase III.

Progress Overview (2 pts)

(Summarize how much of the project is done, what's behind schedule, what's ahead of schedule. Max 200 words.)

- By the end of Phase II, we have completed the initial planning and design of JEEVANROUTE. So far, we have finalized the project architecture, divided the system into modules, and established a shared GitHub repository for team collaboration.
- We have identified the main components of the system, including patient assessment, emergency prioritization, ambulance selection, hospital matching, and route calculation. We have also planned the data structures required to manage patient details, ambulance availability, hospital facilities, and road network information.
- The project design includes the use of priority queues for handling critical emergencies and graphs for representing road networks and calculating efficient routes. We have also outlined how ambulance allocation and hospital selection will work together to meet the patient's medical needs.
- The main work is left which is to develop the different modules that will help the system to select an ambulance, match it with a suitable hospital, and calculate the best route using graphs. In Phase III, we will focus on connecting these parts and check whether the complete system works as expected.
- We did not set separate deadlines for each task, so we cannot clearly say whether any particular task is ahead of schedule or running late.
- Overall, Phase II has helped us understand how to develop JEEVANROUTE and plan the remaining work. In Phase III, we will complete the pending features, connect the modules, and test the system in different emergency situations to see how well it performs.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

- Repository : JEEVANROUTE - GitHub
- Repository Link : https://github.com/Abhishek-co-10/JEEVANROUTE.git
- Branch: Main
- Repository Type: Public
- Source Directory: src

Testing and Validation Status (2 pts)

(Provide information about any tests conducted)

Test Type	Status (Pass/Fail)	Notes
Patient Data Management	Pass	Planned storage and retrieval of patient details.
Emergency Prioritization	In Progress	Priority queue logic planned for critical cases.
Ambulance Availability	In Progress	Ambulance availability and status checking planned.
Hospital Facility Matching	In Progress	Hospital facilities to be checked against patient needs.
Graph-Based Route Calculation	In Progress	

Deliverables Progress (2 pts)

(Summarize the current status of all key project deliverables mentioned earlier. Indicate whether each deliverable is completed, in progress, or pending.)

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